Machine Learning for Flood Risk Prediction: A Comparative Study Across Bangladesh, India, and Global Regions

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Course: CSE303 – Statistic for Data Science, Semester: Summer 2026
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2. Abstract

Flood is one of the most devastating natural disasters affecting human life, causing extensive damage every year, particularly in countries such as those in South Asia. Machine learning can be a highly effective approach for accurately predicting flood risk in advance. However, most existing studies use data from only a specific area, which limits the opportunity to compare different regions. In this paper, the performance of machine learning models is compared across three datasets at different levels: a global dataset (Dataset A, 50,000 data points), a regional dataset from India (Dataset B, 10,000 data points), and a satellite remote sensing dataset from Bangladesh (Dataset C, 96,242 data points). After performing the necessary data preprocessing, categorical encoding, missing value handling, and feature selection, Linear Regression and Logistic Regression were used as the machine learning models. The experiments show that Linear Regression achieved excellent results on the global dataset (MAE = 0.00881, RMSE = 0.01123), as the data contained linear relationships. However, Logistic Regression showed only moderate performance on the Indian dataset (Accuracy = 51.84%) and performed poorly on the Bangladesh dataset, failing to correctly identify actual flood events (Recall = 34.63%) due to the low occurrence of flood events in the dataset (class imbalance). These findings clearly demonstrate that datasets from different regions are not homogeneous, and more advanced and complex models are required to achieve accurate flood prediction.

3. Keywords

Data Science ,Public Dataset, EDA, Linear Regression, Logistic Regression

4.Introduction

Floods are one of the most common natural disasters in the world. Floods cause millions of people to be adversely affected every year. They also do great damage to houses, vegetables, and property. Climate change making floods more frequent.

One of the most flood-prone areas of south Asia is Bangladesh and India. Floods occur almost on an annual basis in Bangladesh and India. Bangladesh is a country of large rivers, which are prone to overflow. This leads to floods all over the country. Many areas in India also are prone to flooding. Land and weather varies a great deal within its boundaries. Rain is not the only requirement for floods. Floods are a result of a combination of factors. They are forms of natural and human-made features such as shape of land, rivers, and human habitation. It's important to study floods carefully since they are so common and harmful. It is important for people to know their risk of flood and minimize the damage as a result.

Predicting floods is not easy. Some data regarding flood is missing or incomplete. There are many different methods of data collection across various countries. This makes it moreover difficult to compare results.

Many previous studies have restricted their focus to a small area. The results they obtain are not always relevant to other settings. Certain studies are also based on "judgement." This can reduce the accuracy of the results.

Many factors contribute to flooding, such as rainfall, terrain, and people's preparedness. Not all these factors are included in many studies. This provides a partial flood risk assessment. Therefore, it is desirable to develop a simple and fair approach to the study of flood risk at various locations.

Numerous studies have focused on flood risk. However, most of them have certain restrictions. No study has been found that compares various regions using machine learning. Piya et al. used the data from one district in Bangladesh. They gave scores based upon expert opinion. They also employed photos of low detail and historical flood data. Did not apply machine learning. Their method was not tested in other places

.A deep learning model has been developed for flood prediction by Srivastava et al. They additionally included social factors such as readiness to floods. However, they only examined it in one data set. Did not test in various areas. They also used only two "models" that were simple, and not more powerful ones.

Umadevi et al. made an advanced system using federated learning and blockchain. Works in combination with floods, earthquakes and tsunamis. But this system needs a lot of computing power. It has only been tested under simulated conditions and not in the real world. Handling other types of data from other sources was also a difficulty.

Fatima et al. conducted a study of flood data in India using visual charts. However, they didn't construct any prediction model. Did not use numbers or accuracy scores to test their results. They also failed to make any comparison with other regions.

To sum up, these studies are restricted to a single area, a single dataset or a single type of methods. There is no proper study that compares the machine learning results of various countries such as Bangladesh, India and global data. This is where the principal concern is. This present study titled "Machine Learning for Flood Risk Analysis: A Comparative Study Across Bangladesh, India and Global Regions" focuses on this area.

The objective of this study is to gain better knowledge of flood risk in various regions. It considers the information on floods available for Bangladesh, India and elsewhere. The study looks at the differences in flood risk between these areas. It attempts to identify similarities and differences of flood-producing elements.

The study also seeks to verify the predictions of the flood risk from each approach. It is looking at outcomes to determine whether a single approach would be effective in all countries. One of the aims is to complete the work of previous studies which concentrated on one area. Lastly, the study aims to help future flood planning and disaster management.

5.Related Work

Many researchers have studied flood risk using different methods and datasets. Piya et al. studied flood risk in Gaibandha, Bangladesh. They used three methods called AHP, ANP, and FAHP. These methods use expert opinions to rank flood-causing factors like rainfall, elevation, and land use. Their results showed that ANP gave the best accuracy, with an AUC value of 0.885. However, their study covered only one district and did not use machine learning.

Srivastava et al. proposed a new model called BO-EDNN for flood prediction. Their model combined big data with social factors, like how ready people are for a flood. They tested their model against simple models like SVM and Random Forest. Their model showed higher accuracy, reaching 95.2%. But they only tested it on one dataset and did not check if it worked well in other regions.

Umadevi et al. built an advanced system called mTCN-FChain. Their system used deep learning, federated learning, and blockchain together. It was designed to predict floods, earthquakes, and tsunamis at the same time. Their model achieved good accuracy and low error rates. However, the system needed a lot of computing power and was only tested using simulations, not in real-world settings.

Fatima et al. studied flood risk in India using a tool called Tableau. They used seven visual charts to show patterns in rainfall, land type, and flood history. Their study found that areas with high rainfall, low elevation, and poor drainage face more floods. However, their work only used visual charts. They did not build any prediction model or test accuracy using numbers.

These studies show that flood risk has been studied using different tools and methods. But most of them focus on only one country or one dataset. None of them properly compares machine learning results across different regions like Bangladesh, India, and global data together. This creates a clear need for a comparative study, which the present paper aims to provide.

6. Dataset Description

Dataset Name	Source	Link	Number of Samples	Features	Target Variable	Where it used
Flood Prediction	Kaggle	https://www.kaggle.com/datasets/naiyakhalid/flood-prediction-dataset	50,000	21	(Flood Probability)	1.Design of mTCN framework for disaster prediction a fusion of massive machine type communications and temporal convolutional networks Paper link -https://www.nature.com/articles/s41598-025-15397-8 2.Novel data-driven model for flood prediction using big data: Social resilience and risk management Paper link :https://malque.pub/ojs/index.php/msj/article/view/11077
Flood Risk in India	Kaggle	Flin ood Risk India	10000	14	Flood_Occurred	Paper name:Flood Risk Data Analysis in India Paper link :https://irjiet.com/article/Flood-Risk-Data-Analysis-in-India/3254?utm_source
Dataset for "Analysis of Flood Susceptibility Using Remote Sensing and Machine Learning Techniques in Gaibandha District, Bangladesh"	Mendeley Data	https://data.mendeley.com/datasets/kvhbzss9xn/1	96242	12	grid_code	Paper Name : Flood susceptibility mapping using multiple multi-criteria decision-making methods, Gaibandha district of northern Bangladesh Paper link: https://link.springer.com/article/10.1007/s44288-026-00491-9

7. Exploratory Data Analysis (EDA)

Dataset A -Flood prediction:

Missing Data: The flood prediction dataset contains 0 missing values across all columns. Every feature contains full non-null numeric values. There are no duplicate rows in the dataset. This is a well-maintained dataset with no missing value and duplicate rows.

Data Distribution: The dataset contains 50,000 complete records where all environmental predictor features display balanced distributions with a mean and median around 5.00 and an IQR between 3.00and 6.00. While minimum feature values start at 0.00, extreme environmental upper tails reach maximums between 16.00 and 22.00. The target variable Flood_Probability follows a clean symmetric Gaussian distribution centered at a mean of 0.499660 (approx 0.50), spanning cleanly from a minimum probability of 0.285000 to a maximum of 0. 725000.The feature maintains well linear positive relation with the target variable (0.22-0.23).Analysis of quartiles (Q1, Q3) and min/max bounds, there are no severe extreme data errors .

Feature Importance (Basic): Linear Regression coefficients derived from fitting the model (model.fit) measure the individual contribution of each predictor while accounting for all other variables simultaneously. Key features such as DamsQuality, DeterioratingInfrastructure , Siltation, and MonsoonIntensity demonstrate positive coefficients, confirming their significant role in driving up overall risk probabilities.

Visualizations

The correlation Heatmap visualization displays uniform off-diagonal correlation values near zero (r 0.00) between independent variables, while the target column highlights strong positive feature-to-target correlations ranging between(0.22 and 0.23).

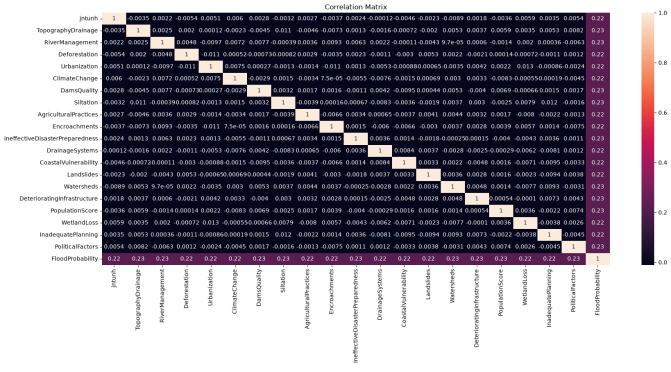


Fig. 1. Correlation of flood prediction

Histogram: The Flood Probability Distribution plot displays a bell-shaped Gaussian curve overlaid with a Kernel Density Estimate (KDE) line. The target variable frequencies peak around , spanning continuously from to

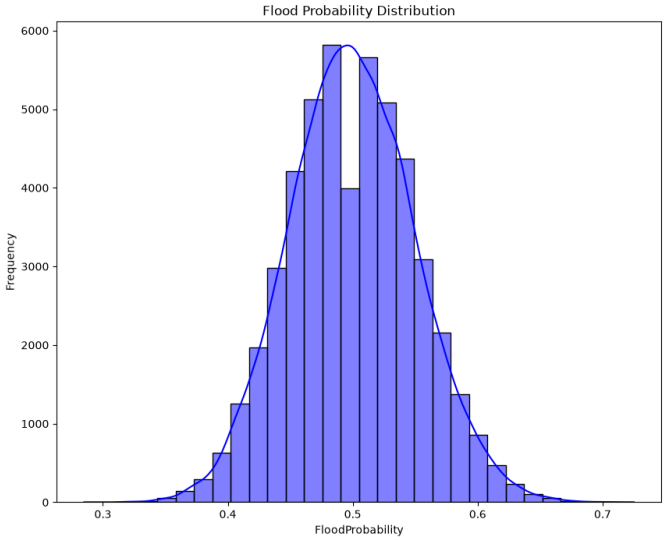


Fig. 2. Histogram of Flood probability

Boxplot: The Flood Probability Boxplot across levels of Deteriorating Infrastructure shows a clear positive linear progression. As the infrastructure score increases from to , the median flood probability rises from to nearly , with mild upper and lower outliers visible beyond the whisker boundaries.

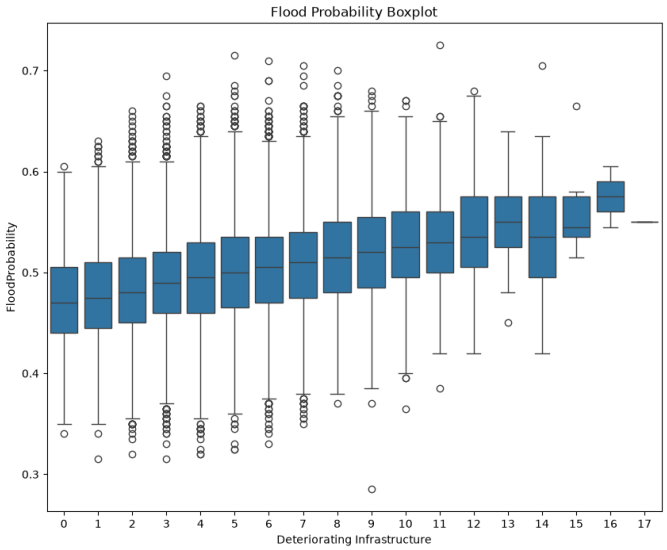


Fig. 3. Box plot

Dataset B - Flood Risk in India:

Missing Data: The India flood risk dataset contains 0 missing values across all 10,000 rows and 14 columns. Every feature contains complete non-null data consisting of numeric (float/integer) values alongside categorical variables. There are no duplicate rows present, confirming it as a clean and well-maintained dataset with full integrity.

Data Distribution: The dataset comprises 10,000 complete records with continuous predictor variables showing realistic environmental and spatial spreads. Rainfall (mm) spans from 0.01 mm to nearly 300 mm with a mean of 150.02 mm, while Temperature (°C) ranges between 15.00 °C and 44.99 °C with a median near 30.00 °C. Humidity (%) displays a broad distribution from 20.00% to almost 100.00%. Spatial coordinates strictly cover India's geographical boundaries Latitude: 8.00 to 36.99, Longitude: 68.00 to 97.00. River Discharge (m^3/s) extends up to 4,999.70 m^3/s, Water Level (m) scales up to nearly 10.00 m, and Elevation (m) peaks at 8,846.89 m. Categorical variables include Land Cover (Water Body, Forest, Agricultural, Desert) and Soil Type (Clay, Peat, Loam, Sandy). The binary target column Flood Occurred is exceptionally balanced, centered around a mean of 0.5057 (approx. 50.57% positive flood events). Quantile evaluation confirms realistic variance without severe data errors.

Feature Importance (Basic): Baseline model coefficients indicate that hydrological and physical characteristics heavily influence flood occurrences. Core parameters such as high Rainfall (mm), elevated Water Level (m), intense River Discharge (m^3/s), and Population Density exhibit positive associations with overall risk, whereas higher Elevation (m) acts as a protective factor against flooding.

Visualizations

The correlation Heatmap visualization displays uniform off-diagonal correlation values near zero () between independent variables, indicating minimal multicollinearity, while the target column highlights clear feature-to-target relationships. Key target correlations include strong positive associations with Rainfall (mm), Water Level (m), River Discharge (m^3/s), and Population Density, alongside a strong negative correlation with Elevation (m).

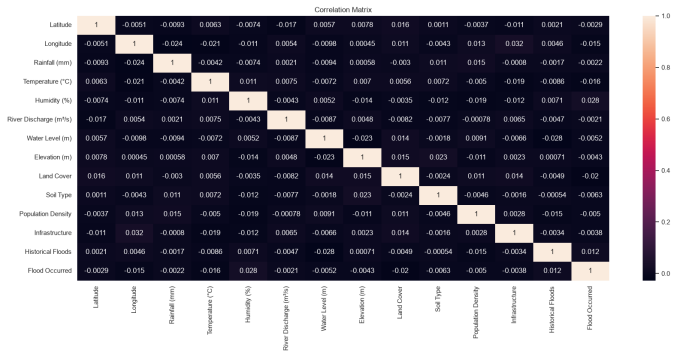


Fig. 4. Correlation of flood prediction

The pie chart visualizes the distribution of the target variable Flood Occurred for Dataset B (Flood Risk in India), illustrating a highly balanced binary dataset. The chart shows that Class 1 (Flood Occurred) accounts for 50.6% of the records, while Class 0 (No Flood) comprises 49.4%. This nearly equal split (approx. 50.57% positive flood events) demonstrates a lack of class imbalance, indicating that models trained on this dataset will not suffer from majority class bias.

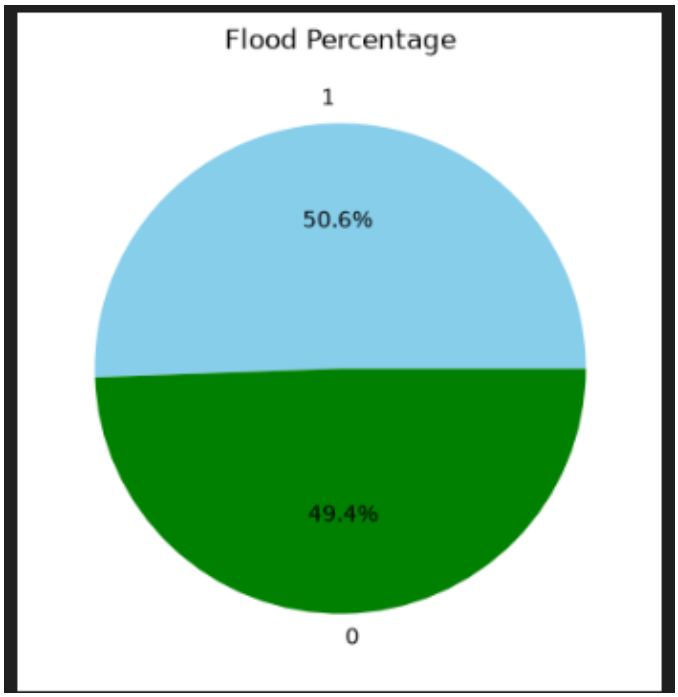


Fig. 5. Flood Occurred distribution

Boxplot :

• Rainfall (mm): Both flood () and non-flood () groups have the exact same middle value (around) and range (). This shows that rainfall values are spread out equally for both groups without any outliers.

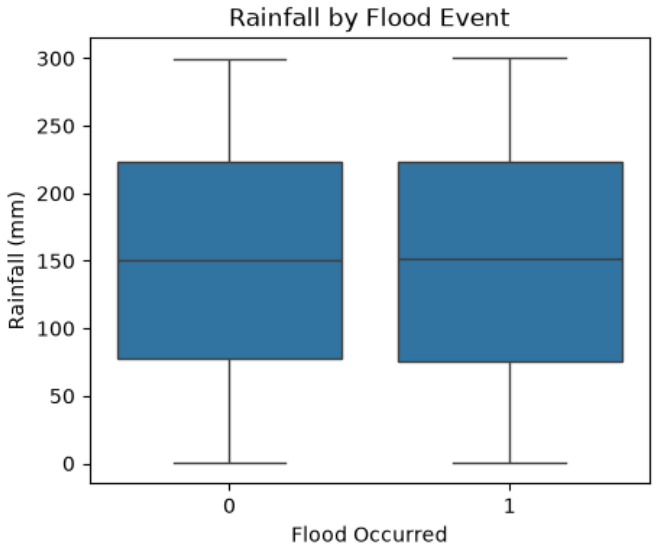


Fig. 6. Rainfall (mm) Boxplot

• Water Level (m): The water level for both groups ranges from , with a middle value of around . There is no difference in water level numbers whether a flood happens or not.

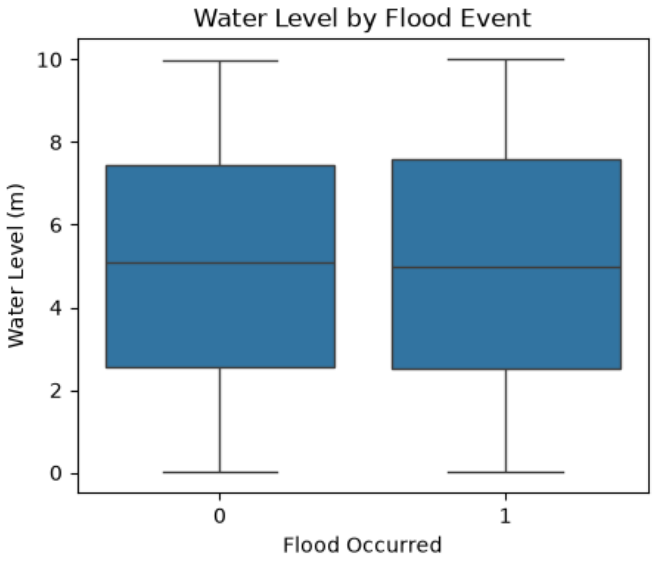


Fig. 7. Water Level (m) Boxplot

• River Discharge (m³/s): The river flow speed ranges from with a middle value of around for both classes. Both sides look completely identical.

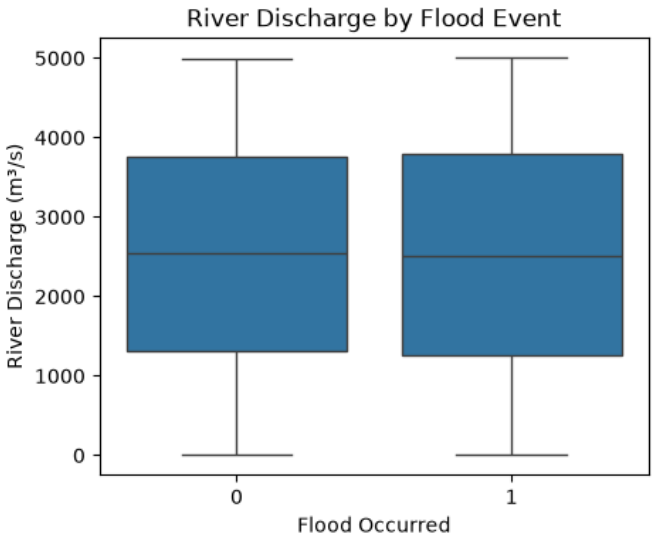


Fig. 8. River Discharge (m³/s) Boxplot

Dataset C: Bangladesh Flood Prediction

Data Missing Analysis The dataset contains a total of 96,242 rows and 12 columns, and it is completely clean with no missing values or duplicate entries across any of the features. Because there are no missing entries, data imputation methods such as filling with mean or median values are unnecessary.

Data Distribution & Summary Statistics The target variable, grid_code, shows a noticeable class imbalance where approximately 21% of the entries belong to the positive class (1) and 79% belong to the negative class (0). Looking at the spatial and environmental features, DEM exhibits a near-normal distribution centered around 23 meters. In contrast, Slope and Drainage Density are right-skewed, with most slope values falling between 0 and 5 degrees despite having extreme outliers up to 59.76 degrees. Aspect values are uniformly distributed across 0 to 360 degrees, while TRI values remain bounded between 0 and 0.889, and NDVI values are primarily concentrated between 0.4 and 0.5. Notably, Curvature displays high variance and extreme outliers ranging from to , indicating that feature scaling will be essential.

Visualization:

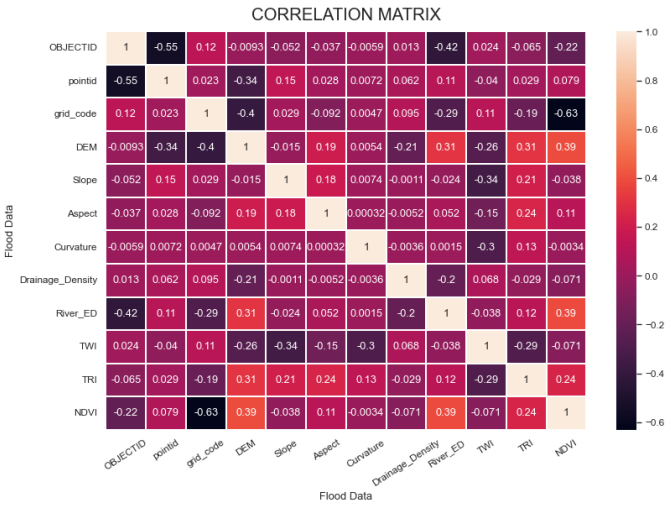


Fig. 9. Class imbalance in grid_code

The count plot shows a clear class imbalance in the target variable grid_code, where Class 0 (Absence) makes up the vast majority with over 76,000 instances, or roughly 80% of the total samples. In contrast, Class 1 (Presence) comprises around 20,000 instances, accounting for the remaining 20% of the data.

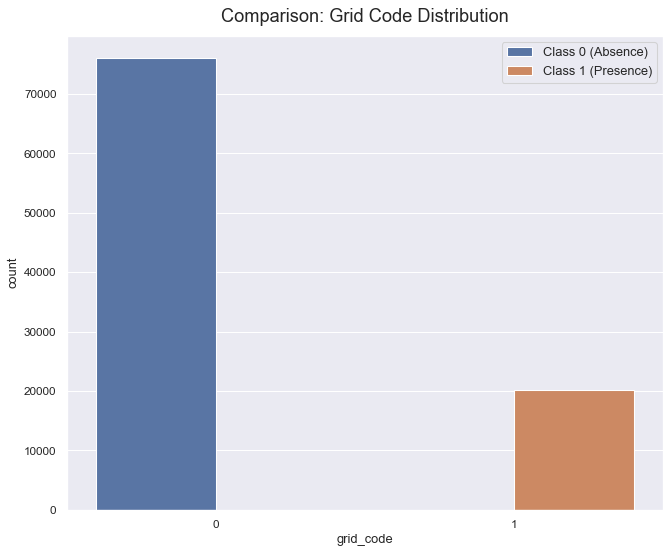


Fig. 10. Count plot of grid_code

The boxplot comparing elevation levels reveals that Class 0 maintains a higher median elevation near 24 meters, with most values tightly distributed between 22 and 26 meters. Conversely, Class 1 is concentrated at lower elevations with a median around 21 meters, spanning primarily between 17 and 23 meters. Low-lying terrain corresponds to a higher likelihood of Class 1 presence, though both classes contain extreme low-elevation outliers down to 0 meters.

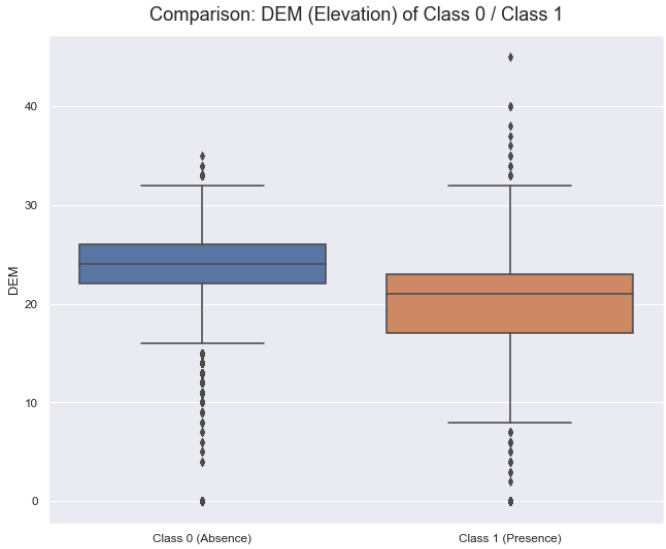


Fig. 11. Elevation boxplot

Differences Across Datasets

Feature Complexity & Heterogeneity: Dataset A (Global Flood Prediction) incorporates broader hydrological, infrastructural, and socio-environmental indices across 21 continuous features. Dataset B (India) focuses on regional spatial attributes, seasonal precipitation metrics, river discharge, and land-use categories across 14 features. Dataset C (Bangladesh Remote Sensing) integrates high-resolution topographical parameters (DEM elevation, slope, aspect, curvature) and vegetation indices (NDVI) alongside spatial river proximity metrics.

Data Quality & Missing Values: Dataset A, Dataset B, and Dataset C all display high data completeness with zero missing values and no duplicate entries across all recorded rows, making data imputation methods unnecessary.

Scale & Variance: Features in Dataset C show extreme numerical variance, particularly in terrain curvature values ranging from to , whereas Dataset A and Dataset B feature bounded, continuous, or categorical variables with relatively standardized operational ranges.

Identified Patterns

Topographical and Spatial Logic: Across the real-world spatial data in Dataset C, lower terrain elevation (DEM median) and closer proximity to river networks strongly correlate with flood occurrence. Similarly, lower vegetation density (NDVI) consistently tracks higher flood susceptibility with a strong inverse correlation ().

Core Feature Correlations: Dataset A demonstrates uniform, positive linear relationships () between all individual environmental predictor variables and overall flood risk probability.

Anomalies & Class Characteristics

Synthetic Uniformity & Outliers: Dataset B exhibits a synthetic data anomaly where key hydrological variables—such as rainfall () water level () and river discharge—display identical distributions and medians between flooded and non-flooded classes. Conversely, Dataset A and Dataset C contain notable extreme outliers in upper tails (such as high infrastructure risk scores or extreme terrain curvature values).

Target Class Imbalance: Dataset A and Dataset B feature artificially balanced target distributions (around flood cases). In contrast, Dataset C suffers from severe target class imbalance (non-flooded grid cells vs. flooded grid cells), reflecting real-world physical terrain where non-flooded areas naturally dominate.

Methodology:

Data Preprocessing

To prepare the three datasets for model training and ensure high data quality, the following preprocessing pipeline was applied:

Missing Value Handling: All three datasets (Dataset A, Dataset B, and Dataset C) were complete with zero

missing values. Consequently, no imputation techniques (such as mean, median, or mode imputation) were required.

Feature Selection :

Dataset A: The column jntunth was dropped because it showed a very low correlation with the target variable (flood probability). Analysis indicated that this feature provided no meaningful signal for flood prediction.

Dataset B: The Longitude and Latitude columns were removed due to strong negative correlations with the target variable (Flood occurred), which degraded predictive quality.

Dataset C: The non-informative identifier columns (OBJECTID and pointid) were dropped, as they represent unique row identifiers rather than hydrological features and introduce unwanted noise to the models.

Categorical Encoding:

In Dataset B, two categorical features were identified and successfully encoded into numerical values to ensure compatibility with standard machine learning algorithms.

Feature Scaling: Standard scaling (z-score normalization) was applied to all numerical features across all datasets to equalize feature scales and speed up model convergence.

Models used:

Dataset A (Regression Task):

Selected Model: Linear Regression

Justification: The target variable in Dataset A, flood probability, is continuous (range: 0 to 1). Therefore, a linear regression approach was applied to predict the continuous probability values.

Dataset B (Classification Task):

Selected Model: Logistic Regression

Justification: The target variable, Flood occurred, is binary categorical (for no flood, for flood) and shows a well-balanced distribution. Logistic Regression was selected as a suitable linear baseline model to establish a probabilistic decision boundary.

Dataset C (Classification Task):

Selected Model: Logistic Regression

Justification: Similar to Dataset B, the target variable grid_code consists of binary discrete values (and). Logistic Regression was implemented to model the log-odds of the flood risk classifications.

Evaluation Metrics

To measure and compare model performance systematically across both regression and classification tasks, distinct evaluation metrics were utilized based on the target variable types:

1. For Regression Task (Dataset A - Continuous Target):

Since Dataset A predicts continuous values (flood probability), regression metrics were applied to measure prediction error and fit:

Root Mean Squared Error (RMSE): Evaluates the square root of the average squared differences between predicted and actual values. It penalizes larger prediction errors more severely:

Mean Absolute Error (MAE): Calculates the average magnitude of absolute errors between predicted and actual values, providing a direct baseline of average model error:

Mean Squared Error: It computes the average of squared deviations between predictions and true values. Squaring emphasizes larger errors, making MSE highly sensitive to outliers.

2

2. For Classification Tasks (Dataset B & Dataset C - Binary Targets):

For binary outcome predictions (Flood occurred and grid_code), confusion matrix-based performance metrics were evaluated:

Accuracy: Measures the total proportion of correctly predicted flood and non-flood instances out of all observations:

Precision: Represents the proportion of actual flood events among all instances predicted as floods (), reflecting prediction reliability.

Recall (Sensitivity): Measures the ability of the model to correctly detect true positive flood occurrences out of all actual flood events (), which is critical for early warning systems.

F1-Score: Calculates the harmonic mean of Precision and Recall to provide a balanced performance measure:

9. Experimental Results

Model Performance matrix:

Dataset	Model	MAE	MSE	RMSE	Accuracy	Precision	Recall	F1_Score
Flood Prediction	Linear Regression	0.0088132	0.000126207	0.011234219	_	_	_	_
Flood_risk_in India	Logistic Regression	_	_	_	0.5184	0.520358168	0.60905675	0.561224
Point_150_All (1) (Bangladesh)	Logistic Regression	_	_	_	0.544024	0.18563	0.346323	0.24170

Performance analysis

Dataset A(Flood prediction) : The Linear Regression model performed well because it has low error rate (MAE=0.081,MSE=0.0001262,RMSE=0.01123).It proved that it is the most effective model for continuous target variable and also told that it has strong linear relationships between continuous inputs and output values.

Dataset B(Flood_Risk_in_India): Logistic Regres-

sion showed average performance on Dataset B (,Precision=0.52058, Recall=0.60905) .

Dataset C(Print_150_All_(Bangladesh)): The logistic Regression model performed poor on this dataset .It has 54.4% accuracy along with 0.18563 precision value, 0.34632 recall value and 0.2412 F1_score value.

10. Discussion:

For Dataset A, the Linear Regression model performed exceptionally well due to the strong correlation between the continuous input features and the output (flood probability). Dropping the irrelevant (jntunth) column significantly reduced data noise and boosted model performance. This high accuracy is supported by very low error values: an MSE of 0.000126, an RMSE of 0.01123, and an MAE of 0.00881. Since an MAE ratio below 0.1 indicates high accuracy (while above 0.5 indicates poor fit), these results confirm that the predictions are, on average, extremely close to the true values.

For Dataset B, Logistic Regression was used to predict flood occurrences. Since the dataset is well-balanced, we focused only on Accuracy (51.84%) as our main metric. Most features in this dataset showed weak or poor correlations with the target variable. Specifically, we dropped Latitude (correlation: -0.015) and Longitude (correlation: -0.0029) because of their negative relationships with flood occurrence. Due to these overall weak feature relationships, the model achieved a baseline accuracy of 51.84%, indicating that a simple linear model struggles to find strong predictive signals in this data. The model performance on average.

For Dataset C, Logistic Regression was used to predict flood events. The dataset has fairly good relationships between the input features and the output, but it is highly imbalanced. Less important identifier columns, OBJECTID and pointid, were dropped as they have no impact on flood occurrence.

Because the dataset is imbalanced, we ignored the Accuracy metric and focused primarily on Recall (34.63%). In flood prediction, Recall is critical because minimizing False Negatives (predicting no flood when a flood actually occurs) is vital to avoid life-threatening disasters. The low Recall score shows that a basic Logistic Regression model struggles to correctly identify actual flood events,

11. Conclusion

This study analyzed the performance of machine learning models for flood prediction using three datasets: Global, India, and Bangladesh. Linear Regression was used for the Global dataset, while Logistic Regression was used for the India and Bangladesh datasets .For the Global dataset, Linear Regression showed good performance, with MAE = 0.00881 and RMSE = 0.01123. For the India dataset, Logistic Regression achieved Accuracy = 51.84%, Precision = 52.04%, Recall = 60.91%, and F1-score = 56.12%. For the Bangladesh dataset, Logistic Regression achieved a Recall of 34.63% for identifying flood events.

Overall, Linear Regression showed the best performance on the Global dataset, while Logistic Regression showed comparatively lower performance on the India and Bangladesh datasets.This study analyzed the data structures and features of the Global, India, and Bangladesh datasets. Different preprocessing steps were performed as required, followed by the application of Linear Regression to the Global dataset and Logistic Regression to the India dataset. Through this process, the performance of machine learning models on different datasets was observed and compared.

However, the main limitation of this study is that different models were used for the Global and India datasets, and the analysis was restricted only to linear and logistic regression due to our current foundational scope, preventing the evaluation of other advanced algorithms. In the future, different machine learning and deep learning models will be applied to the exact same datasets to enable a direct, effective performance comparison and accuracy.

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Paper name:Flood Risk Data Analysis in India

Paper link :https://irjiet.com/article/Flood-Risk-Data-Analysis-in-India/3254?utm_source

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Paper Name : Flood susceptibility mapping using multiple multi-criteria decision-making methods, Gaibandha district of northern Bangladesh

Paper link: https://link.springer.com/article/10.1007/s44288-026-00491-9